Joint Training with Machine-Generated Text Detection Can Distort Sarcasm Detection: Diagnosing Class-Conditional Negative Transfer across Chinese and English

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Abstract

Sarcasm detection requires inferring intent beyond literal wording. A shared encoder jointly trained for sarcasm detection and machine-generated text detection (MGTD) may allow source-predictive cues to influence sarcasm predictions. We harmonize Chinese and English resources, generate topic-matched AI texts, and construct DualTest for source × sarcasm subgroup analysis; AI-side sarcasm labels denote the prompt-specified style rather than independently perceived sarcasm. For AI-generated texts prompted to be non-sarcastic, the evaluated plain joint-training procedure yields a false-positive rate 67.0 percentage points higher than its matched sarcasm-only procedure. Within the AI-generated subset, the ranking induced by sarcasm scores remains largely preserved, indicating a source-conditioned upward score shift rather than collapsed discrimination. The joint procedure also yields the higher false-positive rate in both the Chinese and English strata and on outputs from a second generator under fixed detectors. Representation diagnostics support a source-correlated shortcut hypothesis and identify a constructed high-variance direction to which the subgroup gap is sensitive. In an exploratory downstream analysis, predicted human-authorship probabilities do not consistently track LLM ratings of human-likeness obtained without ground-truth source labels. These findings motivate source × class subgroup evaluation and external validation before detector probabilities are reused as quality scores or rewards.

1 Introduction

Sarcasm is a pragmatic phenomenon in which a speaker's intended attitude may differ from an utterance's literal meaning. Interpreting it relies on implicit contrast, communicative intent, and context, making it difficult to detect automatically. Detection errors can reverse sentiment predictions and disrupt opinion-mining or content-moderation systems (Tay et al., 2018; Vidgen et al., 2019). Annotation outcomes also vary with elicitation method and annotator perspective (Oprea and Magdy, 2020; Casola et al., 2024), while model behavior is sensitive to domain, style, and language (Jang and Frassinelli, 2024; Casola et al., 2024) (§2.1).

One potential way to expand sarcasm resources is to generate additional examples with LLMs. Their usefulness depends on whether they realize the style specified in the prompt without allowing source-specific regularities to become shortcuts. Evaluation must therefore separate sarcasm behavior by source: human-authored and AI-generated texts can each be sarcastic or non-sarcastic, yielding all four combinations.

Source and sarcasm can be predicted by separate models. When both outputs are desired, however, a shared encoder with task-specific heads can reuse one representation rather than duplicate encoders. This convenience may couple the tasks. Machine-generated text detection (MGTD) must distinguish human-written from AI-generated text and thus favors source-predictive representations (Guo et al., 2023; Wang et al., 2024) (§2.2); these stylistic or distributional cues may overlap with features used for sarcasm detection. The labels do not conflict; the potential conflict lies in how the tasks organize a shared representation. MGTD must encode source, whereas sarcasm detection should reflect intent rather than source. Shared-encoder multi-task learning can suffer task conflict and negative transfer by exploiting information useful for one task but spurious for another (Mueller et al., 2022; Hu et al., 2022b). We therefore test whether joint training changes sarcasm predictions.

Source-predictive regularities and sarcasm cues may vary across language and corpus settings (Jang and Frassinelli, 2024; Casola et al., 2024; Wang et al., 2024). We therefore examine this phenomenon in Chinese and English and report language-stratified results to assess the robustness of the subgroup pattern across the two settings.

To test how joint MGTD training changes sarcasm behavior, we compare joint dual-task procedures with matched sarcasm-only controls and build DualTest for source × sarcasm subgroup analysis. AI-side labels denote prompt-specified style rather than independently perceived sarcasm. The evaluated joint-training procedure sharply increases false positives on AI texts prompted to be non-sarcastic; we call this class-conditional negative transfer. Within AI texts, sarcasm-score ranking remains largely preserved, while the joint model exhibits an upward score shift.

We then examine the shared representation. Source information is highly decodable and correlates with sarcasm decisions. To connect encoded information to actual decisions beyond probe accuracy (Hewitt and Liang, 2019; Elazar et al., 2021), we combine probes, task-head correlations, and representation interventions. These analyses identify a constructed high-variance direction to which the subgroup gap is sensitive, supporting a source-correlated shortcut hypothesis and providing a focused target for further causal analysis (§5.3.2, Appendix E). The plain joint procedure also yields a higher false-positive rate than its matched sarcasm-only control on Qwen3.5-generated replacements (§5.3.3).

Detector scores can also become components of generation rewards: ViSP includes a DIP-derived sarcasm score in its PPO reward (Wang et al., 2026; Wen et al., 2023). Evaluating human-like sarcasm generation involves two objectives: conveying sarcasm and being perceived as human-like. A dual-head detector may appear to provide both signals, but human-authorship probability is not a measure of perceived human-likeness. We therefore compare detector scores with LLM ratings of sarcasm and perceived human-likeness obtained without ground-truth source labels in an exploratory analysis (§6).

Contributions. We construct DualTest from Chinese and English resources for source × sarcasm diagnosis of joint MGTD–sarcasm training. Same-architecture comparisons between joint dual-task procedures and matched sarcasm-only controls show class-conditional negative transfer. The degradation is concentrated on AI texts prompted to be non-sarcastic and appears as an upward shift in sarcasm scores. The joint procedure again yields the higher false-positive rate on outputs from a second generator under fixed detectors. Representation diagnostics support a source-correlated shortcut hypothesis and provide a focused target for further causal analysis. An exploratory detector–judge comparison illustrates the risk of treating detector probabilities as measures of sarcasm quality or human-likeness.

2 Related Work

2.1 Sarcasm Detection, Synthetic Data, and Multilingual Resources

Feature-engineered methods (Riloff et al., 2013), neural sequence models (Ghosh and Veale, 2016; Tay et al., 2018), and pretrained language models (Khatri and P, 2020; Du et al., 2022) have each advanced sarcasm detection. Performance nevertheless varies substantially across datasets because label definitions, domains, languages, and modalities differ. Label schemes may reflect intended or annotator-perceived sarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022; Casola et al., 2024), while multimodal resources may use modality-specific annotations (Yue et al., 2024).

Recent work has begun to evaluate LLM-generated sarcasm directly: Majer et al. (2026) evaluate predictive performance, sample diversity, linguistic properties, and human judgments, while Mahato et al. (2026) associate statistical characteristics of synthetic data with shortcut learning and poor transfer to human-authored texts. Motivated by these findings, we examine whether source-predictive regularities learned through MGTD interfere with sarcasm detection when the two tasks share an encoder.

2.2 Machine-Generated Text Detection and Authorship Verification

MGTD methods include zero-shot approaches based on model likelihoods, such as probability-curvature detectors (Mitchell et al., 2023), as well as discriminative models trained on paired human–AI corpora (HC3; Guo et al., 2023). Generalization is a central difficulty: M4 (Wang et al., 2024) provides a benchmark across generators, domains, and languages and shows that detectors struggle to generalize to unseen domains and generators, while RAID (Dugan et al., 2024) stress-tests detectors under unseen generators, decoding variations, and adversarial attacks. This generalization gap is consistent with reliance on source-predictive regularities that are partly generator- or domain-specific; such cues are relevant here because they may enter a representation shared with sarcasm detection. Bevendorff et al. (2025) distinguish closed-set authorship attribution from open-set authorship verification and argue that realistic LLM detection is better formulated as open-set authorship verification. Our binary MGTD setup is narrower: we use it as a controlled source-classification auxiliary task, not as a measure of text quality.

To the best of our knowledge, the interaction between sarcasm detection and MGTD under shared-encoder joint training has not been directly examined. This matters because source is the target of MGTD but should not by itself determine sarcasm predictions; we test whether a shared representation preserves that distinction.

2.3 Multi-Task Negative Transfer, Shortcut Learning, and Shared-Encoder Mitigation

In multi-task learning, auxiliary tasks are not always beneficial: transfer depends on task relatedness and alignment between their data distributions (Zhang et al., 2023; Wu et al., 2020), while gradient conflict is a central optimization challenge that can contribute to negative transfer (Yu et al., 2020; Liu et al., 2021). Mueller et al. (2022) provide a closely related NLP study based on matched single- and multi-task comparisons. They find similar negative-transfer patterns in canonical and text-to-text systems and note that models may infer task identity from input distributions, a strategy that may not transfer reliably out of distribution. He and Choi (2021) and Li et al. (2023) further analyze functional specialization in multi-task Transformer encoders. Shortcut learning offers a complementary perspective: models may exploit features that are predictive in the training distribution but are unreliable under distribution shift (Geirhos et al., 2020; Ye et al., 2026). In MTL, shared representations can increase reliance on spurious correlations (Hu et al., 2022b; Chai et al., 2025). Related multi-dataset work shows that adding data can introduce spurious correlations and reduce worst-group performance (Compton et al., 2023).

Approaches for mitigating interference in shared encoders either isolate task-specific parameters (Pfeiffer et al., 2021, 2020; Karimi Mahabadi et al., 2021; Stickland and Murray, 2019) or modify information flow within the shared computation graph (Wu et al., 2019; Pilault et al., 2021). Our analysis is closer to the second design direction: we compare the plain shared-encoder baseline with two intermediate-representation variants and test how the magnitude of negative transfer varies across source × sarcasm subgroups (§5.3).

2.4 Misalignment in Detector-Based Evaluation and Rewards

The representational conflict discussed above illustrates a broader risk that arises when detector outputs are reused beyond classification, particularly as evaluation or reward signals for generated text. Automated scorers, including LLM judges, are increasingly used to evaluate generated text (Zheng et al., 2023). Generation models are also optimized against model-based reward signals, but overoptimization can improve the proxy score while degrading performance on a gold-standard measure (Gao et al., 2023). A detector can achieve useful classification accuracy without its continuous output being calibrated as a measure of style strength or generation quality. A recent example arises in multimodal sarcasm generation: ViSP (Wang et al., 2026) includes a DIP-derived sarcasm score (Wen et al., 2023) in its PPO reward (Schulman et al., 2017). As an exploratory downstream analysis, Section 6 compares predicted human-authorship probabilities with separately obtained LLM ratings of perceived human-likeness that were produced without ground-truth source labels.

3 Dataset Construction

Testing cross-task interference requires distinct source and sarcasm supervision, with both attributes labeled for evaluation. Here "source" indicates whether a text is human-authored or AI-generated, whereas "corpus" refers to the original dataset from which it was drawn. We combine the English iSarcasmEval data (Oprea and Magdy, 2020; Abu Farha et al., 2022), CSC (Jang and Frassinelli, 2024; Jang et al., 2023), ToSarcasm (Liang et al., 2022), News Headlines v2 (Misra and Arora, 2023; Misra and Grover, 2021), and the Open Chinese Internet Sarcasm Corpus (Zhu, 2022). None provides both a source label and a sarcasm label. Mapping the original annotations to a common binary sarcasm label space yields 44,862 instances (approximately 87% English). Four resources then serve distinct roles: SarcTrain for sarcasm supervision, AuthTrain for source supervision, GenTrain for generation fine-tuning, and DualTest for source $\times$ sarcasm evaluation (Appendix A).

SarcTrain. SarcTrain contains 10,000 human-authored texts for sarcasm classification (2,500 per language $\times$ label bucket). Three buckets are down-sampled; the Chinese sarcastic bucket is expanded from 2,396 to 2,500 by sampling 104 records with replacement. We partition the dataset into five folds. Because oversampling precedes fold assignment, some duplicated records occur in different folds; Appendix A quantifies this dependence.

AuthTrain. AuthTrain contains 19,995 texts for source classification. Seed-level quality control excludes one Chinese SarcTrain instance, leaving 9,999 approximately language-balanced human seeds (5,000 English; 4,999 Chinese). The locally deployed model identified by the Ollama tag `deepseek-r1:8b` (DeepSeek-AI, 2025; Ollama, 2025) generates the AI counterparts. After generation and quality control, 9,996 counterparts are retained; three seeds remain unmatched (99.97% pairing coverage; Appendix A).

We use four language $\times$ style templates, selecting each seed's prompt style from its sarcasm label. Source and prompt-specified style are thus distinct, approximately balanced factors, enabling source $\times$ sarcasm subgroup analysis (Appendix A).

GenTrain. GenTrain contains 2,428 texts for the supervised fine-tuning (SFT) used in §6. From 5,257 eligible records labeled as sarcastic in iSarcasmEval and ToSarcasm, we sample 2,103 (40%; 1,214 English and 889 Chinese). We then sample 325 additional Chinese records with replacement, yielding 1,214 instances per language. By prompt type, the set contains 374 rewriting, 839 response, and 1,215 comment instances (Appendix A).

DualTest. DualTest contains 4,664 texts for source $\times$ sarcasm evaluation. Its human-authored subset combines the official test sets of iSarcasmEval (1,400 English texts) and ToSarcasm (972 Chinese texts). After generation and quality control, 2,292 of the 2,372 human test texts retain AI counterparts (96.6% coverage). Human-authored texts retain their original dataset annotations. For AI-generated texts, the sarcasm label records the prompt-specified style rather than an independently annotated judgment of perceived sarcasm.

The four groups contain 823 human-sarcastic and 1,549 human-non-sarcastic texts, plus 779 AI texts prompted to be sarcastic and 1,513 prompted to be non-sarcastic. Record-level checks find no overlap of SarcTrain or AuthTrain with either side of DualTest; Appendix A reports these checks and the language $\times$ source $\times$ sarcasm distribution.

4 Joint and Single-Task Models

4.1 Tasks and Architecture

Given a Chinese or English input text x, the model predicts a binary sarcasm label and a binary source label; all four label combinations are valid. The tasks share a single DeBERTa-v3-large encoder (He et al., 2023), allowing us to test how joint MGTD–sarcasm training changes the representation read by the sarcasm head. We compare three representation configurations and examine how subgroup errors vary across them; Figure 1 shows the RCNN configuration. We include an RCNN intermediate module because RCNNs combine recurrent context with convolutional pooling for text classification (Lai et al., 2015) and have previously been evaluated for multilingual sarcasm detection (Yang and Ikeda, 2024).

The RCNN configuration applies a single-layer BiLSTM followed by parallel one-dimensional convolutions and global max pooling. With hidden size 256 per direction, the BiLSTM yields a 512-dimensional vector at each token. We concatenate it with the 1,024-dimensional encoder output and apply kernels of widths 3, 4, and 5, each with 128 output channels. Global max pooling yields a 384-dimensional vector z, which passes through dropout to the two heads.

The plain configuration sends `[CLS]` directly to both heads. The task-branch interaction-projection configuration forms two learned 256-dimensional projections, h_s and h_a, and updates them as $h'_s = h_s + P_{a\to s}(h_a)$ and $h'_a = h_a + P_{s\to a}(h_s)$. Because each branch supplies a single key–value vector, all attention weights equal one; the updates are residual cross-branch affine projections, where each P is the composition of the corresponding value and output projections. All retain the same encoder and task definitions (Appendix B).

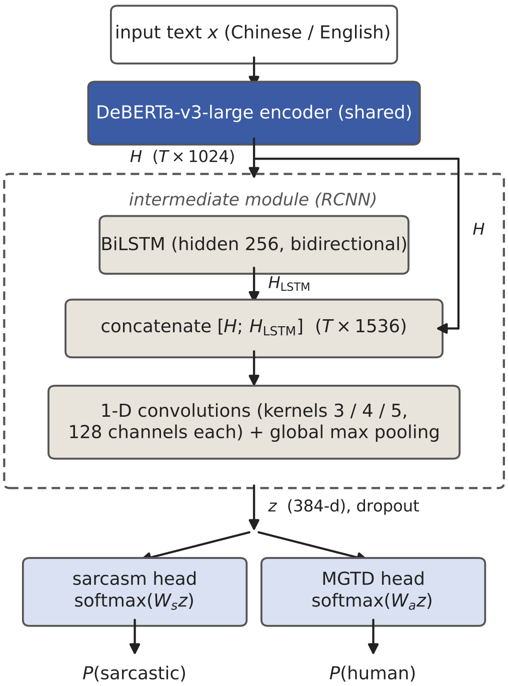

Figure 1: RCNN dual-head configuration: a shared DeBERTa-v3-large encoder, an RCNN intermediate module (dashed), and two task-specific heads. The comparisons in §5 remove this component in the plain configuration or replace it with a task-branch interaction-projection module, while retaining the same encoder backbone and task definitions (Appendix B).

4.2 Training Protocol

We compare joint dual-task and sarcasm-only procedures in the plain and RCNN configurations. Within each architecture, they share the encoder, representation configuration, sarcasm data, folds, and sarcasm-specific hyperparameters. The joint procedure also receives AuthTrain batches, updates shared parameters from both tasks, follows a mixed-task schedule, and uses joint early stopping. Compute exposure differs: the control traverses the full sarcasm fold each epoch, whereas the joint procedure samples sarcasm batches within the mixed schedule. The resulting contrast characterizes the complete joint-training procedure.

Training uses task-homogeneous batches in a 2:1 MGTD:sarcasm ratio, approximating the AuthTrain:SarcTrain size ratio. Let θ_{sh} denote parameters updated by both tasks, let $z_s(x)$ and $z_a(x)$ be the representations read by the sarcasm and MGTD heads, and let W_s and W_a denote the two classification heads. The two representations coincide in the plain and RCNN configurations; in the interaction-projection configuration, they are the task-specific h'_s and h'_a defined above. For each batch B_t, only the active task loss is computed:

$$\mathcal{L}(B_t) = \begin{cases} \frac{1}{|B_t|} \sum_{(x,y)\in B_t} \mathrm{CE}_w(W_s z_s(x), y), & B_t \subset \mathrm{SarcTrain}, \\ \frac{1}{|B_t|} \sum_{(x,y)\in B_t} \mathrm{CE}(W_a z_a(x), y), & B_t \subset \mathrm{AuthTrain}, \end{cases} \tag{1}$$

where CE_w is a pre-specified cost-sensitive cross-entropy for sarcasm (weights are provided in Appendix B). For the class-balanced SarcTrain dataset, the weighted loss assigns a greater penalty to errors on sarcastic examples. Cross-entropy is computed from logits; softmax is applied only for probability-based evaluation. Each task-homogeneous batch updates the active task head and the shared parameters. Unless otherwise stated, binary predictions use a fixed probability threshold of 0.5 rather than a threshold tuned on DualTest.

The encoder uses a lower learning rate than new modules; the sarcasm-head rate is fixed across configurations, with MGTD-head rates in Appendix B.

For dual-task models, checkpoint selection accounts for both tasks. At epoch e, the early-stopping signal is the equally weighted sum of the two validation losses, each normalized by its first-epoch value:

$$\mathcal{L}^{(e)}_{\text{stop}} = \frac{1}{2}\frac{\mathcal{L}^{(e)}_s}{\mathcal{L}^{(1)}_s} + \frac{1}{2}\frac{\mathcal{L}^{(e)}_a}{\mathcal{L}^{(1)}_a}, \tag{2}$$

where $\mathcal{L}^{(e)}_s$ and $\mathcal{L}^{(e)}_a$ are the epoch-e validation losses for the sarcasm and MGTD tasks on their respective validation sets. Normalization reduces the influence of differences in task-specific loss scales and learning dynamics. Single-task controls use their own validation loss with the same five-epoch patience (Appendix B).

We conduct five runs, each using one SarcTrain fold for validation and the remaining four for training; AuthTrain uses a fixed 90/10 split. Each run reloads the pretrained encoder and randomly initializes new modules and heads. Configurations use the same fold assignments and random-seed values; parameter initialization may nevertheless differ across architectures.

DualTest is not included in any training or validation split for either task. Five-run standard deviations reflect joint variation in folds and initialization, which are not separately identified. Wilson intervals are conditional on each fixed final model and quantify test-instance uncertainty, not uncertainty from training or text generation (Table 3; Appendix C.8).

For each configuration used in the subgroup analysis, a final model is trained on all applicable data with a fixed epoch count determined from the five validation runs, an independent seed, and no early stopping. DualTest is not used to train these models or determine their training duration, and the final models are not included in five-run aggregates; §6 uses the final RCNN dual-head model (Appendix B).

5 Experiments and Diagnosis

After training, we use DualTest to compare encoder backbones and representation configurations, evaluate joint- versus single-task procedures, and analyze source × sarcasm subgroups. We report five-run aggregate comparisons and use separately trained final models for paired subgroup analyses and Wilson intervals.

5.1 Backbone Comparison

We compare dual-head mBERT (Devlin et al., 2019), XLM-RoBERTa-base (Conneau et al., 2020), DeBERTa-v3-base, and DeBERTa-v3-large (He et al., 2023) under one evaluation protocol. For DeBERTa-v3-large, we additionally evaluate the interaction-projection and RCNN representation configurations. On the full DualTest set, we report positive-class F1 for the sarcastic class in sarcasm detection and the human-authored class in MGTD.

Table 1 shows higher mean F1 for the DeBERTa configurations than for mBERT and XLM-R. The RCNN configuration achieves the highest mean scores among those evaluated, reaching 0.6309 sarcasm F1 and 0.9301 MGTD F1. The two added representation configurations also show smaller standard deviations across the five runs than the plain large model. Because backbone family, model scale, and representation configuration vary simultaneously, these results permit only configuration-level comparisons. For the separately trained final RCNN model, sarcasm F1 is 0.819 in Chinese and 0.429 in English. Because prevalence, genre, annotation scheme, and language vary together, this comparison does not isolate a language effect. Appendix C provides macro-averaged, human-subset, and language-stratified results.

Model	**Sarc. F1**	**MGTD F1**
mBERT dual-head	0.4945 $\pm$.036	0.8743 $\pm$.006
XLM-RoBERTa-base dual-head	0.4603 $\pm$.057	0.8321 $\pm$.025
DeBERTa-v3-base dual-head	0.5646 $\pm$.056	0.8952 $\pm$.003
DeBERTa-v3-large plain dual-head	0.6126 $\pm$.031	0.9221 $\pm$.010
+ interaction projection	0.6076 $\pm$.020	0.9126 $\pm$.007
+ RCNN	**0.6309** $\pm$.020	**0.9301** $\pm$.005

Table 1: Descriptive dual-head results across backbones and representation configurations (DualTest; five-run mean $\pm$ standard deviation; bold indicates the highest mean).

Arch.	**Task**	**Dual**	**Single**	Δ
Plain	Sarc.	0.6126 $\pm$.031	0.6734 $\pm$.010	$-$**6.08**
Plain	MGTD	0.9221 $\pm$.010	0.9054 $\pm$.027	$+1.67$
RCNN	Sarc.	0.6309 $\pm$.020	0.6751 $\pm$.022	$-$**4.43**
RCNN	MGTD	0.9301 $\pm$.005	0.9253 $\pm$.006	$+0.49$

Table 2: Joint dual-task vs. corresponding single-task procedures (mean $\pm$ standard deviation over five fold-based runs; Δ = dual $-$ single, in percentage points).

5.2 Dual-Head versus Single-Head Comparison

Within each architecture, the joint dual-task and sarcasm-only models share the encoder, representation configuration, sarcasm data, fold assignments, and sarcasm-specific hyperparameters; their task schedules and sarcasm exposure differ as described in §4.2.

Here, negative transfer denotes lower performance than the corresponding single-task procedure. Joint training lowers sarcasm F1 in all ten split-matched comparisons—five per architecture—with differences from -0.8 to -8.6 pp. Mean sarcasm F1 decreases by 6.08 pp for the plain model and 4.43 pp for RCNN, whereas mean MGTD F1 changes by only $+1.67$ and $+0.49$ pp. Sarcasm detection is the only task for which the joint model underperforms its corresponding single-task model in all ten comparisons. The dual–single gap is smaller for RCNN than for the plain architecture. We next examine which source × sarcasm subgroups account for the aggregate degradation.

5.3 Source × Sarcasm Subgroup Analysis

The aggregate F1 decrease is highly uneven across source × sarcasm groups. We therefore examine the four DualTest source × sarcasm subgroups separately. Human-authored texts retain dataset sarcasm annotations; AI-generated labels denote prompt-specified styles. Because each group contains only one reference class, we report recall for sarcastic texts and false-positive rate for non-sarcastic texts.

5.3.1 Three Architectures and Single-Head Controls

The three dual-head configurations characterize how subgroup errors vary across representations. Plain and RCNN single-head models provide same-architecture controls. Within each pair, both models use the same all-human sarcasm data; only the dual-head model is additionally trained on MGTD. The contrast captures the prediction change between the two training procedures. Table 3 reports sarcasm predictions; corresponding MGTD results are in Appendix C.

False-positive rates on AI × non-sarcastic instances are 86.3% for the plain dual-head model, 40.9% for interaction projection, 34.1% for RCNN, and 19.3% for both same-architecture single-head controls; the intermediate rates are therefore more than 45 pp below the plain rate.

On the same AI non-sarcastic texts, 1,021 false positives occur only under the plain dual-head model, versus 7 only under its single-head control ($p < .001$). The large increase occurs under the evaluated joint procedure relative to its sarcasm-only control. RCNN also has a higher false-positive rate under the joint procedure, with a smaller gap. The plain and RCNN models provide same-architecture contrasts, while interaction projection provides a complementary comparison across configurations.

On the same AI × non-sarcastic subgroup, the MGTD head's human-authorship false-positive rate is 6.1% for plain and 6.6% for RCNN. Within AI-generated texts, sarcasm AUC against the prompt-specified style labels remains 0.82–0.88 across models, so the relative ranking of sarcastic and non-sarcastic texts is largely preserved. The joint procedure instead shifts sarcasm probabilities upward, causing many texts prompted to be non-sarcastic to cross the fixed 0.5 threshold. This source-conditioned score shift, rather than a collapse of within-source ranking, characterizes the subgroup failure.

The matched plain comparison makes the concentration especially clear: its non-sarcastic false-positive rate changes by 4.3 pp on human text but by 67.0 pp on AI-generated text.[1] The aggregate F1 decrease is therefore concentrated in a sharply localized source × class subgroup rather than distributed uniformly across DualTest.

5.3.2 Representation-Level Diagnostics

We next ask whether source information in the shared representation is associated with the higher sarcasm scores. A five-fold cross-validated probe on full DualTest representations predicts source from both the joint dual-task (AUC 0.986) and matched sarcasm-only (0.974) models. Source decodability alone therefore does not explain the joint-versus-single-task gap. In the dual-head model, texts with a higher MGTD-head AI logit also tend to have a higher sarcasm logit in the target subgroup ($\rho = 0.519$). After covariance normalization, the probe and MGTD-head source directions have absolute cosine similarities of 0.41 and 0.49 with the sarcasm direction. Together, the logit correlation and directional alignment provide correlational evidence linking source-related structure to the affected sarcasm decisions.

We then modify the frozen representations and recompute predictions. In this separate split-sample analysis, the held-out-half source-probe baseline is 0.982. LEACE (Belrose et al., 2023) lowers it to 0.589 but mainly reallocates errors between source groups. The AI/human non-sarcastic false-positive rates change from 84.8%/32.8% to 68.7%/49.5%, while their unweighted mean remains nearly unchanged (58.8% to 59.1%).

As a split-sample orthogonal-direction control, we remove the maximum-variance fitting-half direction after projecting out the estimated linear source direction. This changes the AI/human false-positive rates to 66.7%/54.1% while measured source decodability remains 0.982; none of the ten sampled random whitened directions produces a comparable change. These controls identify a constructed high-variance direction to which source-group errors are sensitive. Combined with the preceding logit correlations and directional alignment, the results support a source-correlated shortcut hypothesis and identify a candidate representation direction for further causal analysis (Appendix E).

[1] Percentage-point differences are computed from unrounded rates and can therefore differ by 0.1 pp from subtracting the rounded values shown in the tables.

Model	H sarc. recall	H non-sarc. FPR	AI sarc. recall	AI non-sarc. FPR
Plain dual-head	67.1%	32.9%	98.2%	**86.3%** [84.5, 88.0]
Interaction-projection dual-head	71.9%	37.6%	94.4%	**40.9%** [38.5, 43.4]
RCNN dual-head	80.8%	40.5%	89.3%	**34.1%** [31.8, 36.5]
Plain single-head (sarcasm only)	67.8%	28.7%	76.6%	**19.3%** [17.4, 21.4]
RCNN single-head (sarcasm only)	66.2%	28.5%	79.1%	**19.3%** [17.4, 21.4]

Table 3: Sarcasm predictions across four source $\times$ sarcasm groups. Human texts use dataset sarcasm annotations; AI-side sarcasm labels denote prompt-specified styles. Bold marks the focal AI non-sarcastic subgroup. Brackets show Wilson 95% intervals for the fixed final models; full intervals and paired tests are in Appendix C.8.

5.3.3 Fixed-Detector Evaluation on a Second Generator

Using the same four language $\times$ style prompt templates, we generate a Qwen3.5-9B test-side replacement set (Ollama tag `qwen3.5:9b`) (Qwen Team, 2026) while keeping AuthTrain and the detectors fixed. After quality control, the non-sarcastic subsets generated by `deepseek-r1:8b` and Qwen3.5-9B contain 1,513 and 1,550 outputs. The plain joint-versus-single-task gap is 67.0 pp on the DeepSeek-generated subset and 83.4 pp on the Qwen3.5-generated subset. On Qwen3.5 outputs, the RCNN and interaction-projection false-positive rates remain below the plain dual-head rate, at 22.1% and 29.5%. The plain joint procedure again yields a higher false-positive rate than its matched sarcasm-only control, and the three dual-head configurations retain the same ordering by false-positive rate, although the magnitudes differ (Appendix C.8).

6 Exploratory Comparison of Detector Scores with LLM Ratings

As an exploratory downstream analysis, we compare RCNN human-authorship and sarcasm probabilities with sarcasm and human-likeness ratings from DeepSeek-V3 (DeepSeek-AI, 2024), Gemini-2.5-Flash-Lite (Google DeepMind, 2025), and GPT-5-nano (OpenAI, 2025) obtained without ground-truth source labels. We evaluate 600 paired outputs—one from base Qwen3-8B (Yang et al., 2025) and one from its QLoRA-adapted version fine-tuned on GenTrain (Dettmers et al., 2023)—for 300 prompts. Prompts comprise 180 Chinese topic-comment and 120 English rewriting tasks, so language and task are not independently varied. We examine the agreement expected of generation-evaluation proxies.

In Chinese topic-comment generation, mean predicted sarcasm probability rises from 0.780 to 0.895 while the mean 1–5 LLM rating falls from 3.99 to 3.80. Spearman ρ between probability and the mean available rating is -0.020 in Chinese and $+0.338$ in English. Human-authorship probability rises in both settings while human-likeness declines; the pooled ρ is -0.212, but within-condition values of $+0.03$ (base) and -0.12 (SFT) show that opposing condition means dominate.

The predicted-sarcasm change differs between settings ($+0.115$ for Chinese topic-comment; -0.124 for English rewriting). Across six comparisons formed by two settings and three judges, fine-tuned outputs receive lower mean sarcasm ratings in every case; five remain significant after BH-FDR correction. Classification probabilities therefore need not provide the text-level ranking or condition-level direction expected of a style or quality proxy.

Pairwise judge rank correlations are low to moderate, so the ratings provide an external comparison signal rather than a gold standard. The plain detector changes the corresponding correlations by at most 0.02. These disagreements motivate external validation before detector probabilities are reused as quality scores or rewards. Appendix C gives the protocol and tests; Appendix D provides selected cases.

7 Conclusion

Our results identify source-conditioned negative transfer under the evaluated joint MGTD–sarcasm procedure. On AI texts prompted to be non-sarcastic, the plain joint false-positive rate against prompt-specified style labels is 67.0 pp above its matched sarcasm-only control. Ranking within AI texts remains largely preserved, indicating an upward shift rather than collapsed discrimination. The joint procedure yields the higher false-positive rate in both language strata and with fixed detectors on Qwen3.5-9B outputs; RCNN and interaction projection have lower focal FPRs among evaluated configurations. Diagnostics support a source-correlated shortcut hypothesis and identify a candidate representation direction. Human-authorship probabilities diverge from human-likeness ratings, motivating external validation.

Limitations

DualTest combines human sarcasm annotations with prompt-specified style labels for AI-generated texts. Accordingly, AI-side false positives are defined relative to the style specified in the prompt rather than independent perceived-sarcasm annotation.

The core joint-versus-sarcasm-only comparisons evaluate the complete joint-training procedure. Within each pair, the architecture, sarcasm data, fold assignments, sarcasm-head learning rate, and class weights are held fixed, while MGTD batches, mixed-task scheduling, shared-parameter updates, and joint early stopping are introduced together. Compute- and update-matched controls could help disentangle these components. Cross-architecture results support only configuration-level comparisons because model capacity and optimization differ. The cost-sensitive sarcasm loss may also affect absolute recall and false-positive rates.

Because oversampling precedes fold assignment, some duplicated SarcTrain records occur in different folds and may appear on both sides of a training–validation split, introducing dependence into checkpoint selection. These records do not occur in DualTest. Generation failures from `deepseek-r1:8b` leave 80 human test texts without retained AI counterparts.

The representation analyses on one fixed final model connect source-related structure to behavior but do not isolate a unique causal feature or characterize variation across training seeds. The detector–judge study uses LLM rather than human ratings, and language is not varied independently of generation task. Future work should test these findings using independent style annotations, matched training budgets, human ratings, and evaluation across a broader range of generators.

Ethical considerations

All human-authored texts were obtained from publicly released research datasets: iSarcasmEval, CSC, ToSarcasm, News Headlines, and the Open Chinese Internet Sarcasm Corpus. The iSarcasmEval and CSC repositories display MIT licenses; for sources without an identified redistribution license, we do not redistribute human-authored text (Appendix A). We collected no new user data and recruited no new human participants. We did not perform an additional audit of the released corpora for personally identifiable information or offensive content.

All AI texts were generated locally using publicly released open-weight models (`deepseek-r1:8b`, Qwen3.5-9B, and the base and QLoRA-adapted Qwen3-8B). They were used only for training and evaluation and were labeled as AI-generated. For automated evaluation, evaluation inputs, generated outputs, and rating instructions were transmitted to the DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano APIs. Dataset metadata and ground-truth source labels were not transmitted.

The detectors were developed for research diagnosis. Their outputs should not be used as evidence of authorship or misconduct for individual texts without human review and domain-specific validation. They should not be deployed for content moderation or automated sanctions without evaluating out-of-distribution behavior. The class-conditional failure diagnosed here is an example of systematic subgroup error affecting a particular type of generated content.

AI-assisted tools supported language editing, structural revision, and formatting, but did not generate the reported experimental outputs or statistics. The authors determined the study design, analyses, and conclusions and take full responsibility for the work.

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A Detailed Dataset Construction, Generation, and Quality Checks

Corpus	Lang.	Inst.	Resource use
iSarcasmEval (SemEval-2022 T6)	English	3,467	SarcTrain; GenTrain; DualTest*
CSC	English	6,878	SarcTrain
ToSarcasm	Chinese	3,898	SarcTrain; GenTrain; DualTest*
News Headlines v2	English	28,619	SarcTrain
Open Chinese Internet Sarcasm Corpus	Chinese	2,000	SarcTrain

Table 4: Input corpora and resource use. The instance counts refer to the 44,862-record pooled collection; *DualTest uses separate official test splits excluded from these counts. Genre / annotation: iSarcasmEval — tweets, speaker self-annotation; CSC — conversational responses; ToSarcasm — topic comments, third-party annotation; News Headlines — distant supervision (The Onion / HuffPost); Open Chinese Corpus — Weibo, third-party annotation.

A.1 Source corpora, sampling, and release scope. The pooled collection contains 44,862 instances in total (86.9% English and 13.1% Chinese). SarcTrain is balanced across the four language × label groups by downsampling the two English groups and the Chinese non-sarcastic group, then sampling 104 additional Chinese sarcastic records with replacement. The Chinese sarcastic source pool contains 2,396 records corresponding to 2,394 distinct stored text values; after expansion to 2,500 records, its text-duplication rate is 4.24% (A.3).

The iSarcasmEval and CSC repositories display MIT licenses. We did not identify explicit redistribution licenses in the releases we accessed for ToSarcasm, News Headlines v2, or the Open Chinese Internet Sarcasm Corpus. We therefore do not redistribute human-authored text from those sources. We plan to release only eligible AI-generated text, labels and metadata, and reconstruction scripts. The scripts require users to obtain the human-authored corpora from their original sources.

Lang.	Source	Sarc.	Non-sarc.
English	Human	200	1,200
English	AI	192	1,174
Chinese	Human	623	349
Chinese	AI	587	339

Table 5: Language × source × sarcasm distribution of DualTest. Human texts use dataset sarcasm annotations; AI-side labels denote prompt-specified generation styles rather than independently judged sarcasm.

A.2 Language × source × sarcasm distribution of DualTest.

A.3 Data quality checks. Automated checks confirm language and label balance across the five SarcTrain folds. AuthTrain contains 9,996 complete human–AI pairs and three unmatched human instances. Record-level matching finds that neither SarcTrain nor AuthTrain shares records with either the human or AI side of DualTest.

SarcTrain's text-duplication rate is 0.34% for English and 2.16% for Chinese. Chinese duplication is concentrated in the sarcastic group (4.24%; non-sarcastic group: 0.08%) and results primarily from the oversampling used to fill that group (A.1). Because oversampling precedes fold assignment, for 86 of the 104 added records, another copy originating from the same source record is assigned to a different fold. The two counting rules are summarized below.

Counting rule	Groups	Records	Cross-fold
Exact stored text; oversampled copies included	116	241	98
Case/whitespace normalized; added copies excluded	19	46	16

The first rule underlies the reported duplication rates; the second isolates pre-existing duplicates after the added copies are removed. These relationships are relevant to fold-based checkpoint selection but do not place the affected training or validation records into DualTest. Post-processing detects and removes chain-of-thought-style markers (Wei et al., 2022) and the reasoning prefaces described in A.4.

A.4 AI generation protocol. The AI texts in AuthTrain and the two DualTest versions are generated by locally deployed models through the Ollama API. The model deployed under the Ollama tag `deepseek-r1:8b` generates AuthTrain and the DeepSeek version of DualTest's AI side; Qwen3.5-9B, deployed under the tag `qwen3.5:9b`, generates the test-side replacement set used for fixed-detector evaluation on a second generator. Both models use temperature $= 0.8$, top-p $= 0.95$ (Holtzman et al., 2020), and a maximum output length of 500 tokens. Generation instructions use four templates defined by language $\times$ style, with style determined by the seed instance's sarcasm label. The templates constrain output style and length (20–80 Chinese characters; 10–50 English words). These constraints are intended to reduce, but do not eliminate, source differences in length; Appendix E separately repeats the representation correlation analysis within length tertiles. The complete Chinese sarcastic template used in the experiment is reproduced verbatim below:

```
【任务】生成一条讽刺风格的评论
【主题】{seed}
【要求】
- 直接给出简短的评论，不要解释思路或分析
- 风格要讽刺、幽默、夸张
- 表达与主题相关的观点即可，不必追求严谨
- 20-80 字为佳
【评论】
```

Its English translation is:

```
[Task] Generate a sarcastic comment
[Topic] {seed}
[Requirements]
- Directly provide a short comment;
do not explain your reasoning or
analysis
- The comment should be sarcastic,
humorous, and exaggerated
- It is sufficient to express a
viewpoint related to the topic; the
comment need not be rigorous
- Preferably 20--80 Chinese
characters
[Comment]
```

The Chinese non-sarcastic template differs only in the style line, which requests a normal, natural style. The English templates use the same field structure and request either a sarcastic, humorous, exaggerated style or a normal, natural style, with a length limit of 10–50 words.

Post-processing removes <think>…</think> segments, reasoning prefaces such as "好的" (*hǎo de* 'okay'), "嗯" (*ń* 'hmm'), "Okay," "Let me," and "First," and Markdown markup. Outputs that still exceed the length limit or exhibit cyclic repetition after repeated generation attempts are discarded. AuthTrain retains 9,996 AI counterparts from 9,999 seeds; DualTest retains 2,292 counterparts from 2,372 seeds.

The Qwen3.5-9B test-side replacement set has no generation failures, yielding 1,550 AI $\times$ non-sarcastic instances rather than 1,513. Generator-specific rates therefore use the full set of retained outputs for each generator rather than an identical seed intersection. For this run, Ollama thinking mode is disabled (`think = false`) to prevent the reasoning process from exhausting the output budget. The same post-processing removes any inline chain-of-thought from the `deepseek-r1:8b` outputs.

A.5 SFT training and generation-evaluation set. After eligibility filtering, 5,257 records labeled as sarcastic remain for generation-task construction (2,972 English; 2,285 Chinese). After shuffling with seed 42, we sample 40% of these records, yielding 2,103 instances (1,214 English; 889 Chinese); language balancing then samples 325 additional Chinese records with replacement, producing GenTrain (2,428; 1,214 per language). By prompt type, it comprises 374 rewriting instances, 839 response instances, and 1,215 comment instances. Generation evaluation uses a separately constructed set of 300 prompts that is not part of the GenTrain training–validation split (180 Chinese and 120 English): the Chinese task is to generate a sarcastic comment on a given topic, and the English task is to rewrite a normal sentence as a sarcastic one. Language and generation task are therefore not independently varied. This set is used for detector scoring and for all three LLM evaluations in §6, which are conducted without ground-truth source labels.

B Hyperparameters and Training Details

B.1 Discriminative learning rates. The sarcasm-head learning rate is fixed at 5×10^{-6} in every configuration with an active sarcasm head. The MGTD-head rate is 1×10^{-5} for the plain dual-head model and the plain and RCNN MGTD-only controls; it is 5×10^{-6} for the RCNN and interaction-projection dual-head models. Thus, the core sarcasm dual–single comparisons retain the same sarcasm-specific rates within each architecture. The RCNN dual–single comparison for MGTD also changes the MGTD-head learning rate, so it is reported descriptively rather than as an optimization-matched estimate of transfer; cross-configuration differences can likewise

Configuration	Enc.	Inter.	Sarc.	MGTD
Plain dual-head	5×10^{-7}	—	5×10^{-6}	1×10^{-5}
Plain sarcasm-only	5×10^{-7}	—	5×10^{-6}	—
Plain MGTD-only	5×10^{-7}	—	—	1×10^{-5}
Interaction-projection dual-head	5×10^{-7}	5×10^{-6}	5×10^{-6}	5×10^{-6}
RCNN dual-head	5×10^{-7}	5×10^{-6}	5×10^{-6}	5×10^{-6}
RCNN sarcasm-only	5×10^{-7}	5×10^{-6}	5×10^{-6}	—
RCNN MGTD-only	5×10^{-7}	5×10^{-6}	—	1×10^{-5}

Table 6: Discriminative learning rates for the dual-head and task-specific controls. The plain setting is also used in the backbone comparison. A dash denotes an inactive module or task head.

reflect optimization settings. Early stopping uses the normalized joint validation loss of Equation 2 for dual-head models and the task's own validation loss for the task-specific controls. Both procedures use an early-stopping patience of five epochs.

B.2 Shared training hyperparameters. Training uses `torch.optim.AdamW` with default PyTorch betas $(0.9, 0.999)$ and $\epsilon = 10^{-8}$, weight decay 0.01, and parameter-group learning rates in B.1. We implement a linear 200-step warmup followed by a constant learning rate using PyTorch's `LambdaLR`. Training runs for at most 10 epochs, with an early-stopping patience of 5 and gradient clipping at 1.0. The physical batch size is 4 with gradient accumulation over 4 steps, giving an effective batch size of 16 for the active task. The MGTD:sarcasm sampling ratio is 2:1, and the sarcasm class weights are $[1.0, 2.5]$ for the non-sarcastic and sarcastic classes, respectively. The task ratio and class weights were specified before evaluation on DualTest and were not tuned on DualTest. For the class-balanced SarcTrain dataset, the class weights implement a cost-sensitive objective that assigns a greater penalty to errors on sarcastic examples. The normalized joint validation loss is $0.5\times$ [sarcasm validation loss / first-epoch value] $+0.5\times$ [MGTD validation loss / first-epoch value]. The single-head controls process the full sarcasm training fold each epoch, whereas the dual-head models draw sarcasm batches under the 2:1 task-mixing schedule.

Detector inputs use a maximum sequence length of 512 tokens. In the RCNN module, a width-k convolution uses padding $\lfloor k/2 \rfloor$ followed by ReLU. Global max pooling covers all convolution positions; no post-convolution padding mask is applied.

All fold-based runs use a seed equal to 42 plus the fold index and share the same fold assignments and random-seed values across architectures and ablations. The same fixed AuthTrain 90/10 split is reused in all five runs; only the SarcTrain validation fold changes. For each final-model configuration, the fixed epoch count is the mode of its five best epoch counts plus one, capped at 10 epochs; the resulting counts range from 3 to 10 across configurations. The final model uses all applicable training data, a seed offset of $+1000$, no validation set, and no early stopping. Main tables report five-run means and standard deviations rather than final-model point estimates.

B.3 Model sizes and computational environment. The encoders have approximately 178M (mBERT), 278M (XLM-RoBERTa-base), 184M (DeBERTa-v3-base), and 435M (DeBERTa-v3-large) parameters; the SFT generator Qwen3-8B has about 8B parameters, of which QLoRA updates only low-rank adapters. Detector training and inference are performed on a single NVIDIA GeForce RTX 3090 (24 GB). Across the five-fold backbone, architecture, and ablation experiments and final-model training, total compute is on the order of tens of GPU-hours. A single detector run takes from a few minutes to a few hours, and SFT takes about two hours. The software environment is PyTorch 2.9.0 (CUDA 12.8), transformers 4.57.1, and peft 0.18.0. AI texts are generated locally through Ollama (Appendix A.4); LLM evaluation without ground-truth source labels uses API calls made in May 2026 (Appendix C.3).

B.4 SFT (QLoRA). We fine-tune Qwen3-8B using QLoRA (Dettmers et al., 2023), with 4-bit loading and bfloat16 computation. The LoRA configuration (Hu et al., 2022a) uses rank 64, alpha 16, and dropout 0.0 and targets `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. The maximum sequence length is 2,048, with gradient checkpointing enabled. Training uses a learning rate of 5×10^{-5}, three epochs, batch size 1, gradient accumulation 8, and a warmup ratio of 0.05. After language balancing, a seed-42 split holds out 15% of GenTrain for validation; validation loss selects the checkpoint. Because language-balancing oversampling precedes the split, duplicated Chinese records may occur in both partitions.

C Complete Experimental Results

C.1 Macro-averaged F1 (supplement to Table 1). Under the same five-run protocol, the RCNN macro-averaged F1 scores computed from each run's confusion matrix are **0.6678 ± 0.036** for sarcasm and **0.9298 ± 0.005** for MGTD. Sarcasm macro-F1 is higher than its positive-class F1 (0.6309) because the non-sarcastic majority class has a higher F1. These values supplement the positive-class results in §5.1; because macro-F1 is not reported for every configuration, they are not used for cross-configuration ranking.

C.2 Complete four-quadrant results for the MGTD head (including MGTD-only controls). The reporting convention matches Table 3 (recall for human subsets; false-positive rate for AI subsets).

Model	H×s	H×n	AI×s	AI×n
Plain dual-head	91.4	91.5	3.6	6.1
Interaction-projection dual-head	85.4	85.4	1.8	4.1
RCNN dual-head	94.8	93.2	5.5	6.6
Plain MGTD-only	90.4	86.6	4.1	4.2
RCNN MGTD-only	90.5	92.4	3.6	5.7

Table 7: MGTD-head four-quadrant results (%): recall for human (H) subsets, false-positive rate for AI subsets. s = sarcastic, n = non-sarcastic.

In every subgroup, each MGTD-only control differs from its corresponding dual-head model by fewer than ten percentage points. No analogous extreme class-conditional error appears on the MGTD side, consistent with §5.3.

With the detectors held fixed, the `qwen3.5:9b` replacement set also shows no comparable MGTD subgroup increase: AI × non-sarcastic false-positive rates are 0.9%, 1.3%, and 0.5% for plain, interaction projection, and RCNN; AI × sarcastic rates are at most 0.4%; and overall MGTD F1 scores are 0.952, 0.917, and 0.965.

C.3 Evaluation protocol, inter-judge rank correlations, and group-level tests (§6). Each sample is evaluated separately by each of the three judge APIs in one API call using the same prompt structure for all model conditions (May 2026; temperature $= 0.3$; GPT-5-nano uses its default decoding because it does not support this parameter). The prompt contains only the input and evaluated text, not the ground-truth source label. It asks for source inference followed by 1–5 ratings of sarcasm and human-likeness. The ratings are therefore obtained without ground-truth source labels, although the judges are explicitly asked to infer source before assigning the two ratings. Base and SFT outputs are evaluated separately.

Pairwise inter-judge rank correlations are low to moderate. Spearman ρ ranges from 0.20 to 0.38 for sarcasm (all $p < 10^{-6}$) and from 0.03 to 0.23 for human-likeness (one pair is not significant). The conclusion in §6 therefore rests on consistent condition-level directions; individual-text rankings vary across judges.

Outputs from the SFT model receive lower mean sarcasm ratings in all six language × judge groups; Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995) show that five differences remain significant. The pooled correlations in Table 8 use the three-judge mean as a descriptive aggregate; because pairwise rank correlations are limited, the judge-specific results remain necessary. Four instances each have one unparsable rating and retain two valid judge ratings; the aggregate uses the mean of the available ratings and therefore retains all 600 outputs.

Subset	n	Sarc. ρ	Human-like ρ
Overall	600	$+0.136^*$	$-\mathbf{0.212}^*$
Chinese	360	$-\mathbf{0.020}$	-0.218^*
English	240	$+0.338^*$	-0.270^*

Table 8: Spearman correlations between detector probabilities and LLM-judge ratings (RCNN detector, $n = 600$ pooled over base and SFT): Sarc. ρ compares $P(\text{sarcastic})$ with sarcasm ratings, and Human-like ρ compares $P(\text{human})$ with human-likeness ratings. * significant; the other entries are not significant.

C.4 Paired tests between SFT outputs and human-authored reference texts (supplementary). These tests are performed on the Chinese subset of the generation-evaluation set ($n = 180$; three GPT-5-nano ratings are missing, leaving 177). The comparisons provide judge- and dimension-specific comparison context: for DeepSeek-V3, SFT outputs receive significantly lower human-likeness ratings than the reference texts (-0.161, $p = 0.0025$), whereas for GPT-5-nano, SFT outputs receive significantly higher sarcasm ratings than the reference texts ($+0.186$,

Judge / lang.	n	Base	SFT	adj. p
DeepSeek-V3 / en	120	4.025	3.292	<.0001
DeepSeek-V3 / zh	180	4.044	3.878	.0008
Gemini-2.5-FL / en	120	3.992	3.750	.0008
Gemini-2.5-FL / zh	180	4.028	3.994	.2008*
GPT-5-nano / en	119	4.084	3.681	<.0001
GPT-5-nano / zh	180	3.911	3.517	<.0001

Table 9: Group-level tests for sarcasm ratings obtained without ground-truth source labels (Wilcoxon + BH-FDR). * not significant. GPT-5-nano lacks one English base rating (C.3: $n = 119$) and three Chinese reference ratings (affecting C.4, not the C.3 Chinese row); tests use pairwise deletion.

$p = 0.0114$). The remaining four comparisons do not reach significance at the reported threshold. Taken together, the results indicate that proximity to the human-authored reference texts depends on the judge and evaluated dimension.

C.5 Plain-detector control. With the plain detector, the Chinese sarcasm probability changes from 0.765 to 0.763 (-0.001), the English sarcasm probability changes from 0.843 to 0.659 (-0.184), and predicted human-authorship probability rises by $+0.760$ in Chinese and $+0.705$ in English. The two detectors therefore agree on the rise in predicted human-authorship probability, while the sarcasm direction depends on the detector.

Correlations between detector probabilities and corresponding ratings are $+0.142$ for sarcasm and -0.214 for human-likeness overall; -0.037 (not significant) and -0.209 in Chinese; and $+0.342$ and -0.283 in English. Each differs from the RCNN-detector value by at most 0.02. Negative associations between predicted human-authorship probabilities and human-likeness ratings are similar across the two detectors, whereas the condition-level sarcasm-probability change differs between the plain and RCNN detectors, especially in Chinese.

C.6 Within-condition correlations (RCNN detector). For predicted human-authorship probability and human-likeness ratings, the base condition has $\rho = +0.03$ ($n = 300$, not significant; Chinese $+0.10$, English -0.09). The SFT condition has $\rho = -0.12$ ($p = 0.039$; Chinese -0.19, $p = 0.009$; English -0.12, not significant). The pooled value of -0.212 is driven mainly by the opposing condition-level changes: SFT raises predicted human-authorship probability by about 0.67 while lowering the mean rating. The pooled negative correlation should therefore not be interpreted as a strong instance-level inverse relationship. Within either condition, predicted probability of human authorship is not significantly positively correlated with rated human-likeness.

For sarcasm, within-condition correlations are $+0.05$ for Chinese base (not significant), $+0.15$ for Chinese SFT ($p = 0.048$), $+0.41$ for English base, and $+0.37$ for English SFT (both English values significant). Chinese alignment is therefore at most weakly positive, consistent with its near-zero pooled result.

C.7 Detector probabilities before and after SFT across the two task–language settings. The RCNN detector assigns the following mean probabilities to base and SFT outputs:

Lang.	Cond.	$P(\textbf{sarcastic})$	$P(\textbf{human})$
Chinese	Base → SFT	0.780 → 0.895	0.289 → 0.969
English	Base → SFT	0.854 → 0.730	0.198 → 0.864

Table 10: Mean detector probabilities before and after SFT (RCNN detector).

Sarcasm probabilities move in opposite directions in the Chinese comment-generation and English rewriting settings: $+0.115$ and -0.124, respectively. GenTrain itself is balanced at 1,214 instances per language; the evaluation settings instead differ in task, prompt format, source data, and sample size, all of which may contribute to the contrast. Predicted human-authorship probabilities rise sharply in both settings.

The mean LLM ratings for the human-authored reference texts are 3.77 for sarcasm and 4.17 for human-likeness; the corresponding values are 4.01 and 4.24 for the base model and 3.71 and 4.08 for SFT. From the base model to SFT, the mean human-likeness rating decreases even as predicted human-authorship probability increases, as analyzed in §6.

C.8 Conditional test-instance uncertainty: confidence intervals, paired tests, and subset F1. Conditional on each fixed final model, we quantify test-instance uncertainty with Wilson 95% intervals for quadrant rates and 95% percentile intervals from 10,000 instance-level bootstrap samples for F1. The instance-level bootstrap resamples text instances independently and does not preserve human–AI seed pairs. These intervals do not include uncertainty from training, generator sampling, corpus construction, or human–AI pairing. They complement the between-run standard deviations in Table 2 and Table 1. All values use the final model for each architecture and follow the reporting convention of Table 3.

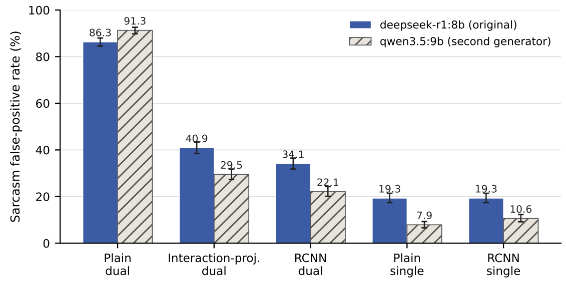

Figure 2: Sarcasm false-positive rates on the AI × non-sarcastic subset for five detectors and two generators. Error bars are Wilson 95% intervals conditional on each fixed model.

For both matched model pairs, the false-positive rate remains higher under the dual-head procedure on the Qwen3.5-generated AI texts, although the magnitudes differ.

Model	DeepSeek	Qwen3.5
Plain dual-head	86.3 [84.5, 88.0]	91.3 [89.8, 92.6]
Interaction-projection dual-head	40.9 [38.5, 43.4]	29.5 [27.3, 31.8]
RCNN dual-head	34.1 [31.8, 36.5]	22.1 [20.1, 24.3]
Plain single-head	19.3 [17.4, 21.4]	7.9 [6.6, 9.3]
RCNN single-head	19.3 [17.4, 21.4]	10.6 [9.2, 12.3]

Table 11: Wilson 95% intervals for the sarcasm false-positive rate on the AI × non-sarcastic subset (two generators; %).

Paired McNemar tests use predictions on the same DeepSeek-generated AI × non-sarcastic texts. The discordant-pair counts are 1,021 / 7 for plain dual vs. plain single, 239 / 15 for RCNN dual vs. RCNN single, 805 / 15 for plain dual vs. RCNN dual, and 203 / 100 for interaction projection vs. RCNN dual (all $p < .001$). The discordant counts and absolute rate differences provide the substantive effect-size information.

In both architectures and both language strata, the false-positive rate is higher under the dual-head procedure than under its single-head control. Language remains coupled with corpus, genre, and annotation setting, so the stratified results do not isolate a language effect.

The final RCNN dual-head model's MGTD F1 is 0.938 [0.931, 0.945] on the full set, 0.961 [0.952, 0.969] in Chinese, and 0.923 [0.912, 0.933] in English. Because the human-only subset contains no AI-authored examples, an MGTD F1 score on that subset would not measure source discrimination; Table 7 therefore reports human recall directly.

Chinese sarcasm F1 is higher than English sarcasm F1 for the RCNN model (0.819 vs. 0.429). This contrast also reflects differences in prevalence, genre, and annotation procedure: sarcasm prevalence is 63.8% in Chinese but 14.2% in English, and the English human side is dominated by author-labeled intended-sarcasm tweets from iSarcasmEval whereas the Chinese side uses topic-comment-style ToSarcasm. The positive-class F1 values should therefore not be interpreted as an isolated language effect.

The separately retrained final RCNN model's overall sarcasm F1 is 0.663, compared with 0.629 on human texts alone, because the overall score also includes AI texts. The five-run mean in Table 1 is a different estimate. Within the human subset, absolute dual–single differences in sarcasm F1 are below 0.03 and vary in direction. The negative-transfer effect is therefore concentrated in the AI-side quadrants, consistent with §5.3.

False-positive rates use a threshold of 0.5. For the plain dual-head model on AI × non-sarcastic text, the median sarcasm probability is 0.767 with an interquartile range of 0.635–0.839. The distribution therefore lies mostly above the threshold rather than placing only a few cases near it. Single-head medians are 0.204 for the plain model and 0.069 for RCNN.

AUC values within AI texts are similar across models (0.82–0.88). Joint training thus raises sarcasm scores while largely preserving their internal ordering. This pattern is consistent with a source-correlated upward shift in sarcasm scores and motivates the representation analyses in Appendix E.

D Illustrative Cases of Disagreement between Detector Probabilities and LLM Ratings

Illustrative cases were manually selected after inspecting the paired table of RCNN-detector probabilities and ratings from the three LLMs to show large disagreements; no formal ranking or top-k selection was performed. The selection is nonsystematic, and the cases should not be used to esti-

Model	Lang.	n	Dual FP; rate [CI]	Single FP; rate [CI]	Δ pp
Plain	en	1,174	987; 84.1 [81.9, 86.1]	242; 20.6 [18.4, 23.0]	+63.5
Plain	zh	339	319; 94.1 [91.1, 96.1]	50; 14.7 [11.4, 18.9]	+79.4
RCNN	en	1,174	445; 37.9 [35.2, 40.7]	247; 21.0 [18.8, 23.5]	+16.9
RCNN	zh	339	71; 20.9 [17.0, 25.6]	45; 13.3 [10.1, 17.3]	+7.7

Table 12: Language-stratified sarcasm false-positive rates on AI texts prompted to be non-sarcastic (%; Wilson 95% intervals). Dual FP and Single FP give the number of false positives under the dual- and single-head models. Exact paired McNemar tests give $p < .001$ for both plain comparisons and RCNN English, and $p = 2.4 \times 10^{-5}$ for RCNN Chinese.

Model	Full set	Chinese	English	Human subset	Human×zh	Human×en
Plain dual-head	0.556	0.746	0.312	0.586	0.710	0.374
Interaction-projection dual-head	0.643	0.781	0.420	0.593	0.739	0.366
RCNN dual-head	0.663 [.646, .680]	0.819 [.802, .836]	0.429 [.399, .458]	0.629 [.605, .653]	0.774 [.748, .798]	0.399 [.355, .440]
Plain single-head	0.661	0.750	0.514	0.612	0.719	0.434
RCNN single-head	0.664	0.758	0.511	0.603	0.709	0.434

Table 13: Per-language and human-subset sarcasm F1 (final models; bootstrap 95% intervals included for RCNN).

mate the prevalence of any disagreement type. Detector probabilities range from 0 to 1, and LLM ratings are averaged across the three evaluators on a 1–5 scale. Long texts are shortened in the table display. Chinese examples retain the original model input and provide an English translation in brackets.

The three groups show, respectively, high predicted human-authorship probabilities with low LLM human-likeness ratings, high predicted sarcasm probabilities with low LLM sarcasm ratings, and the reverse sarcasm disagreement. In the first human-likeness case, AI-generated text with an obvious reasoning-style preface is nevertheless assigned predicted human-authorship probability 1.00 by the MGTD head, corresponding to the disagreement discussed in §6.

Language	Model condition	Detector probability	Mean LLM rating	Text (excerpt)
Chinese	SFT	1.00	2.67	“好，我需要针对三星回怼华为的新闻生成一条讽刺评论。首先……” [“Okay, I need to generate a sarcastic comment on the news of Samsung hitting back at Huawei. First…”] (reasoning-style preface)
Chinese	SFT	1.00	2.33	“跑分竞赛：参数焦虑：跑分焦虑：水货：哈哈哈” [“Benchmark race: spec anxiety: benchmark anxiety: grey-market goods: hahaha”] (degenerate output)
English	SFT	1.00	2.67	“Same as me, I’m not sure” (vacuous short sentence)

Table 14: Cases with high predicted human-authorship probability but low human-likeness ratings. The detector column reports $P(\text{human})$.

Together, these cases motivate candidate hy-

Language	Model condition	Detector probability	Mean LLM rating	Text (excerpt)
Chinese	SFT	1.00	2.67	“三星：我才是第一！你看看我这摄像头总分，109+96，112+89，我赢了！” [“Samsung: I’m the real number one! Look at my camera total scores, 109+96, 112+89, I win!”] (numbers repeated without sarcastic transformation)
Chinese	SFT	0.99	2.67	“恭喜恭喜，美国牛牛牛，继续加油” [“Congratulations, congratulations, America is awesome awesome awesome, keep it up”] (formulaic mock-praise)
English	SFT	0.93	2.67	“2 hours left to tip it over, I’m going to the toilet”

Table 15: Cases with high predicted sarcasm probability but low LLM sarcasm ratings. The detector column reports $P(\text{sarcastic})$.

Language	Model condition	Detector probability	Mean LLM rating	Text (excerpt)
Chinese	Base	0.14	4.00	“特朗普承诺 120 亿补贴像撒大饼，农民收到的却是缩水成硬币的承诺……” [“Trump promised a 12-billion subsidy as if tossing out big pancakes, yet what farmers receive are promises shrunk into coins…”] (metaphorical irony)
Chinese	Base	0.24	4.00	“特斯拉：价格跳水后又跳水，消费者：钱包跳水后又跳水。” [“Tesla: prices dive and then dive again; consumers: wallets dive and then dive again.”] (parallel-structure irony)
English	Base	0.12	4.00	“Thank you so much for this *beloved* review, Mandy! I’m *overjoyed* you enjoyed it” (asterisk emphasis)

Table 16: Cases with low predicted sarcasm probability but high LLM sarcasm ratings. The detector column reports $P(\text{sarcastic})$.

potheses involving surface markers such as mock-praise formulas, exclamation marks, and colloquial short sentences, as well as semantic reversal. Their prevalence and causal effects require systematic testing. The examples illustrate several disagreement types compatible with the near-zero aggregate Chinese sarcasm correlation in §6.

E Representation-Level Analysis of Source-Conditioned Error Allocation

This appendix provides additional protocol details, diagnostic results, and implementation checks for the representation-level tests in §5.3.2. The conclusions are interpreted as described in §5.3.2.

E.1 Setup. All analyses use the frozen plain dual-head model that produces the 86.3% result reported in the main text. We use split-sample evaluation: one half is used to estimate the direction, and the other is used to compute metrics. The appendix baseline is therefore 84.8% on the evaluation half rather than 86.3% on the full set.

The implementation checks yielded the expected numerical behavior. Manually computed logits match model outputs with a maximum error of 1.4×10^{-6}, and the erasure-projection residual is about 10^{-6}. LEACE satisfies its defining property: after erasure, the fitting-half class-mean difference has a relative residual of 7×10^{-6}, and an in-sample linear probe is at chance (AUC 0.514). This in-sample result serves only as an implementation check; held-out post-erasure probe performance is reported in Table 17. The source-probe AUC is 0.986 with true labels, and a permuted-label sanity check gives 0.512. The probe values refer to different evaluation sets: 0.986 is the five-fold result on the unmodified full representation set; 0.982 is the corresponding baseline on the intervention-evaluation half; 0.514 is the in-sample LEACE implementation check; and 0.589 is held-out post-erasure probe performance.

E.2 Protocol. The linear probe is a logistic-regression classifier evaluated with five-fold cross-validation. As a sanity check, a permuted-label control retrains the same probe after shuffling source labels, and correlation analyses are repeated within text-length tertiles. For alignment, Σ is the Ledoit–Wolf shrinkage covariance estimator (Ledoit and Wolf, 2004) fitted to all pooled Dual-Test representations, with eigenvalues clipped below 10^{-8}; w denotes a task-head weight difference or probe direction. The whitening transform is transductive: its covariance is estimated from unlabeled representations in both halves. Whitened alignment is $\cos(\Sigma^{1/2}w_1, \Sigma^{1/2}w_2)$. We report absolute cosine because the sign of a probe direction depends on the source-label convention.

Erasure tests cover one direction, INLP subspaces (k = 1–16; Ravfogel et al., 2020), and LEACE (Belrose et al., 2023). Controls use ten sampled random whitened directions and directions estimated from permuted labels. We construct source-direction-orthogonal PC1 (srcORTH-PC1) by projecting the fitting-half representations onto the orthogonal complement of the estimated linear source direction and then extracting the maximum-variance direction in that projected space. Erasure removes each representation's scalar projection onto this direction. This operation leaves the measured linear source decodability unchanged, although orthogonality to one estimated direction does not imply independence from all multidimensional or nonlinear source information.

E.3 Diagnostic evidence. Source is highly linearly decodable: AUC is 0.986 overall, 0.990 on sarcastic texts, and 0.985 on non-sarcastic texts. The permuted-label control gives 0.512. The single-head model also yields an AUC of 0.974, showing that source decodability is a background property shared by both models. On AI $\times$ non-sarcastic texts, the MGTD-head AI logit and sarcasm-head logit correlate at $\rho = 0.519$; correlations by text-length tertile are 0.44, 0.56, and 0.63.

The sarcasm and source directions are nearly orthogonal in the original geometry ($\cos \approx -0.03$). After covariance normalization, the absolute cosine is 0.41 for the probe-estimated source direction and 0.49 for the MGTD-head source direction. These results provide correlational evidence consistent with the shortcut hypothesis and motivate the intervention tests below.

E.4 Intervention evidence. gap = AI $\times$ non-sarcastic false-positive rate $-$ human $\times$ non-sarcastic false-positive rate; source AUC measures the performance of a probe retrained after erasure.

The summarized single-direction and INLP rows do not show an AI-side reduction comparable to LEACE. LEACE reduces measured linear source decodability and the source-specific gap from 52.0 to 19.1. Its gap change is not matched

Erasure scheme	AI×n FPR	H×n FPR	Gap	AUC
None (baseline)	84.8	32.8	52.0	0.982
Single dir. (head / probe / random)	84.8–86.2	—	—	≈0.98
INLP ($k = 16$)	89.8	—	—	0.974
LEACE (linear)	68.7	49.5	19.1	**0.589**
Random whitened ×10	84.9	32.7	—	0.982
srcORTH-PC1	**66.7**	54.1	12.6	**0.982**

Table 17: Selected intervention summary. H×n = human × non-sarcastic; AI×n = AI × non-sarcastic; FPR in %; AUC is the source-probe AUC after erasure. Bold marks the two diagnostic interventions discussed in §5.3.2, not best performance.

by any of the ten sampled random whitened directions.

The data-derived srcORTH-PC1 diagnostic qualifies this interpretation. It leaves the measured linear source decodability intact but changes the AI-side rate to 66.7 and the source gap to 12.6, an effect similar to LEACE. This effect is not matched by any of the ten sampled random directions. The intervention identifies a candidate direction whose semantic content and role in the training-time transfer pattern remain targets for further causal analysis.

We use an unweighted group mean to give the two non-sarcastic source groups equal importance despite their different sizes. Under LEACE, this mean false-positive rate is almost unchanged (58.8 $\rightarrow$ 59.1). This result shows that the affected directions shape how non-sarcastic errors are allocated between AI and human texts.